

		$107 = 20\pi \sqrt{\frac{r}{a}}$
8	D	$F \sin 30^\circ = m (3 + 5 \sin 30^\circ) \omega^2 \quad \text{--- (1)}$ $F \cos 30^\circ = mg \quad \text{--- (2)}$ $\frac{(1)}{(2)} : \quad \tan 30^\circ = \frac{(3 + 5 \sin 30^\circ) 4\pi^2}{(9.81)T^2}$ $\rightarrow \quad T = 6.2 \text{ s}$
9	C	The gravitational field at point D due to both masses is zero.